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VESUVIUS CHALLENGE: INK DETECTION

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5th June 2026

ABSTRACT

The Vesuvius Challenge is a global open-science initiative designed to decipher carbonized ancient papyri using X-ray micro-tomography. Despite advancements in machine learning, the field remains constrained by a critical scarcity of high-fidelity volumetric training data. This project explores a methodology to address this bottleneck by first developing a robust pipeline for 2D ink detection, a prerequisite for future 3D synthetic data generation. We establish a rigorous mathematical baseline using adaptive Sauvola thresholding and directional morphological filtering to benchmark our primary solution: a deep learning framework leveraging the frozen RADIO Vision Foundation Model (VFM) backbone paired with a custom MLP segmentation head. Evaluated via a Leave-One-Out Cross-Validation (LOOCV) protocol on a pool of high-resolution historical fragments, the fine-tuned RADIO model achieves a Mean IoU of 81.44% and a Recall of 97.72%, significantly outperforming the mathematical baseline (25.15% Mean IoU and 29.43% Recall). This framework was deployed to generate a large-scale, high-fidelity ground truth dataset from the Duke Papyrus Archive and the Oslo collection. By providing this foundational data, this work resolves a significant masking bottleneck, laying the groundwork for future 3D volumetric synthesis and broader epigraphic research within the scientific community.

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CHAPTER 1

INTRODUCTION

1.1 CONTEXT

In 79 AD, the eruption of Mount Vesuvius buried the library of Herculaneum, carbonizing hundreds of ancient papyrus scrolls. Rediscovered in the 18th century, these artifacts survive as densely rolled volumes where the individual layers of papyrus have completely fused together, rendering them impossible to physically unroll without total destruction. Furthermore, ancient carbon ink does not merely sit on the surface but penetrates deeply into the papyrus fiber structure. This intricate 3D morphological signature necessitates the use of high-resolution X-ray micro-tomography (CT scans) rather than simple 2D surface imaging. To decipher these lost texts using this 3D volumetric data, the Vesuvius Challenge was launched as a global, open-science machine learning initiative (Vesuvius Challenge 2023).

The dataset provided by the Challenge is structurally dichotomous. On one hand, there are massive, intact, and entirely unlabelled 3D tomographic volumes of the rolled scrolls. On the other hand, there are small, physically detached flat fragments. These detached fragments can be optically photographed using infrared imaging to reveal the hidden carbon ink, providing the rare and critical 2D ground-truth labels required to train predictive models on the 3D volumes.

1.2 MOTIVATION

Our initial ambition in joining the Vesuvius Challenge was to directly improve the State-of-the-Art (SOTA) 3D ink detection models. However, while benchmarking these architectures, we identified a critical limiting factor: a severe scarcity of labeled, high-fidelity volumetric training data. To overcome this bottleneck, we defined a two-step methodology to artificially expand the data pool: first, masking real ink on uncarbonized historical papyri in 2D, and second, computationally transforming these 2D masks into synthetic 3D volumes that imitate the morphological signature of the Vesuvius CT scans, while preserving authentic human handwriting.

Before developing complex 3D generators, we initiated preliminary exploratory experiments by integrating our basic 2D masks directly into the Challenge’s 3D volumes via mean and max padding techniques. These approaches yielded suboptimal results, confirming that the physical behavior of carbonized ink is highly complex and that simplistic 2D-to-3D projections are fundamentally insufficient.

More importantly, these early trials revealed that the prerequisite first step—extracting 2D reference masks from heavily degraded historical manuscripts—was a formidable computer vision bottleneck in its own right. Because flawed 2D inputs would inevitably corrupt any downstream 3D synthesis, we realized that solving this 2D masking challenge was critical.

1.3 GOALS



FIGURE 1.1

3D tomographic scan of the carbonized scroll PHer. 172 (Scroll 5) alongside virtually unwrapped segments revealing hidden Greek characters. Image reproduced from the Vesuvius Challenge (Vesuvius Challenge 2023).

Consequently, the primary objective of this semester project definitively pivoted toward establishing a scientifically rigorous, highly robust 2D masking pipeline.

Specifically, this report aims to explore how Vision Foundation Models (VFMs) can supersede complex, hand-crafted mathematical filters (Otsu 1979; Sauvola and Pietikäinen 2000) when operating in extreme low-signal-to-noise domains. By fine-tuning a dense visual descriptor architecture, our goal is to accurately isolate faint, degraded ink strokes while remaining impervious to upstream physical damages such as canvas cracks and hallucinated plant fibers.

Ultimately, the overarching goal of this project is to deploy this finalized framework to automatically generate large-scale, high-fidelity 2D ground truth masks from extensive historical archives, such as the Duke Papyrus Archive (David M. Rubenstein Rare Book & Manuscript Library 2026). By definitively solving the 2D masking bottleneck and open-sourcing the resulting dataset, we aim to provide foundational data that not only accelerates future 3D synthetic generation algorithms for the Vesuvius Challenge, but also empowers the broader computational paleography and digital humanities communities.

Chapter Summary: Motivated by the severe scarcity of 3D training data in the Vesuvius Challenge, this project initially aimed to develop a synthetic 3D generator. After identifying the extraction of clean 2D inputs as a major bottleneck, the project strategically pivoted. The new goal is to replace fragile mathematical filters with a robust Vision Foundation Model to construct and publish a massive, high-fidelity 2D dataset, serving both future 3D synthesis and the broader scientific community.

CHAPTER 2

METHODS

To definitively solve the foundational 2D masking bottleneck and provide high-fidelity training data for the Vesuvius Challenge, our methodology evaluates two fundamentally distinct computer vision paradigms. We first establish a highly optimized, handcrafted classical pipeline utilizing local statistical properties to serve as a rigorous mathematical baseline. We then introduce our primary solution: an advanced deep learning framework leveraging a frozen Vision Foundation Model (VFM) backbone paired with a custom Multi-Layer Perceptron (MLP) segmentation head. This dual-paradigm approach is designed to systematically overcome the severe illumination variances, structural noise, and physical degradation that characterize ancient manuscript substrates.

2.1 ADVANCED MATHEMATICAL BASELINE

Global image binarization techniques, such as Otsu’s method (Otsu 1979), invariably fail when applied to historical papyri due to non-uniform aging, ink fading, and severe intra-fragment illumination variance. To establish a robust classical benchmark, we designed a multi-step pipeline utilizing local statistical thresholds and directional morphological filtering.

Prior to adaptive filtering, large physical holes and torn edges within the papyrus are isolated using a strict high-intensity binary threshold. This hole mask is heavily dilated (using a 45×45 kernel) to create a safety buffer that prevents edge artifacts from corrupting subsequent steps.

The pipeline then applies local adaptive Sauvola thresholding (Sauvola and Pietikäinen 2000) to isolate low-contrast regions from the background. The local threshold $T(x, y)$ is computed within a localized sliding window size of 51×51 pixels according to the following formulation:

$$T(x, y) = m(x, y) \cdot \left[1 + k \cdot \left(\frac{s(x, y)}{R} - 1 \right) \right] \quad (2.1)$$

where $m(x, y)$ and $s(x, y)$ represent the local mean and standard deviation within the window, respectively, R is the dynamic range of the standard deviation (set to 128 for 8-bit images), and the control parameter is empirically optimized to $k = 0.25$.

While Sauvola thresholding successfully captures faint ink regions, it introduces extensive background noise by misinterpreting the prominent horizontal carbonized plant fibers of the papyrus substrate as text strokes. To eliminate these directional artifacts, a morphological "scalpel" consisting of an opening operation with a strict directional 1×35 horizontal structural element is applied. To counteract any structural damage caused by this aggressive filtering, a subsequent 2×3 morphological closing operation is executed to topologically reconnect fractured strokes.

Following this, geometrical Connected Components (CC) are extracted. Noise components are filtered out using strict paleographical constraints, discarding any component with a pixel area outside the $60 \leq A \leq 1500$ range or exhibiting an unnatural aspect ratio greater than 3.5. Finally, a global horizontal text-line projection profile is computed across the fragment matrix; any remaining components whose centroids fall outside the detected high-density text valleys are discarded as random speckling.

2.2 VISION FOUNDATION MODEL FEATURE EXTRACTION

For our primary deep learning architecture, we utilize the frozen `NVlabs/RADIO` (Robust Artificial visual Descriptor via feature fusion, version `v2.5-1`) foundation model (Ranzinger et al. 2024). This massive vision transformer (ViT) architecture excels in low-signal-to-noise domains by generating dense, multi-resolution visual descriptors that inherently fuse high-level semantic shapes (such as ancient Greek character morphology) with low-level textural properties (distinguishing genuine carbon ink granules from uninked carbonized plant fibers).

2.2.1 MACROSCOPIC PREPROCESSING AND PATCHING

Prior to slicing the high-resolution papyrus scans, a macroscopic illumination correction step is executed. Contrast Limited Adaptive Histogram Equalization (CLAHE) (Zuiderveld 1994) is applied exclusively to the luminance channel of the RGB images. This targeted normalization preserves the original color metrics and ink hues while significantly enhancing local structural contrast along faded character boundaries.

Following CLAHE normalization, large-scale fragments are sectioned into standardized, non-overlapping input patches of 512×512 pixels. To prevent highly fragmented or entirely empty substrate regions from training the network, a dynamic filtering step evaluates the corresponding physical gap mask; any patch where the vacuum/gap ratio exceeds 95% of the total spatial area is automatically discarded from the dataset.

2.2.2 UPSCALING AND TOKEN GEOMETRY MAPPING

Because ancient ink granules and stroke variations are microscopic, passing standard-resolution patches through a standard Vision Transformer results in a severe loss of spatial detail. To preserve these critical pixel-level features, each 512×512 patch is upscaled by a factor of 4 to a resolution of 2048×2048 pixels via bicubic interpolation before entering the transformer layers.

The ViT backbone processes this enlarged input into a standardized 128×128 grid of spatial tokens. By enforcing the 4x upscaling pre-processing step, each individual token in the resulting transformer layer maps precisely to a localized 4×4 pixel area of the original papyrus fragment. This spatial mapping enables the extraction of hyper-localized micro-features, yielding a dense, high-dimensional feature embedding tensor of size $[4096, 128, 128]$ per patch.

2.3 DATA OPTIMIZATION AND AUGMENTATION PIPELINE

Training a neural network on ancient manuscripts introduces a severe class imbalance, as genuine text occupies a minor fraction of the expansive, empty papyrus substrate. To optimize computational efficiency and prevent the network from converging toward a trivial majority-class solution, we implement a robust offline extraction database utilizing a customized Hard Negative Mining strategy during HDF5 file creation.

While 100% of the dense embedding tokens containing valid, manually verified ink structures are preserved, the remaining negative tokens are stochastically downsampled to enforce a balanced data

distribution. Pure uninked papyrus substrate tokens are restricted to a 35% retention probability, while pure background vacuum or gap tokens are limited to a 15% retention probability.

To ensure the model generalizes perfectly across varying institutional digitization setups and illumination anomalies, the database is augmented offline during the extraction pass. Instead of merely labeling transformations, specific deterministic pixel-level adjustments are mathematically applied: brightness scaling factors of 1.2, 0.8, and a severe 0.5 darkening, alongside intense contrast enhancement factors of 1.2 and 2.0. This comprehensive pipeline artificially expands the morphological variance of the dataset while maintaining strict token alignment with the ground truth labels.

Dataset Partitioning Strategy: To rigorously validate the architecture before large-scale deployment, the total available dataset of 10 annotated fragments was partitioned. A core validation subset of 7 structurally diverse fragments was isolated specifically for the comparative benchmarking and cross-validation (LOOCV) phase. The remaining 3 fragments were withheld from this initial evaluation to serve as a supplementary variance pool, ultimately integrated directly into the final production model to maximize semantic robustness.

2.4 MLP SEGMENTATION HEAD AND HYBRID LOSS OPTIMIZATION

To maintain a lightweight parameter footprint and prevent catastrophic forgetting within the foundation model, the core weights of the RADIO transformer are kept strictly frozen. The extracted 4096-dimensional embedding tokens are flattened and fed into a localized, custom Multilayer Perceptron (MLP) segmentation head.

The architecture of the MLP compresses the high-dimensional feature tokens into a hidden layer of 256 units, regulated by a 0.4 Dropout probability to prevent overfitting on specific handwriting styles. The network then projects the features into a single logit mapping, which reconstructs a $[1, 128, 128]$ probability space corresponding directly to the original token grid layout.

To aggressively penalize the accidental erasure of faint or highly damaged text structures, the network is optimized using a specialized hybrid loss function. This formulation couples a heavily weighted Binary Cross-Entropy (BCE) loss with a structural Dice loss component:

$$\mathcal{L}_{\text{total}}(\theta) = \text{BCE}_{\text{weighted}}(y, \hat{y}, \alpha = 50) + 1.5 \cdot \mathcal{L}_{\text{Dice}}(y, \hat{y}) \quad (2.2)$$

The positive weight multiplier ($\alpha = 50$) forces the gradient descent algorithm to prioritize the retention of sparse text pixels, effectively recovering faded ink strokes that classical filters interpret as background substrate.

2.5 SLIDING WINDOW INFERENCE AND HYBRID POST-PROCESSING

During the production inference phase, full-resolution manuscripts are processed using a sliding window technique with a 50% spatial overlap ($\text{stride} = 256$ pixels) to guarantee smooth transitions. Individual patch predictions are accumulated and averaged across a global matrix to completely eliminate boundary grid artifacts. To transform the raw probability mappings into binary ground truth masks suitable for open-source publication, the accumulated logits undergo a targeted hybrid post-processing sequence:

1. **Edge Artifact Suppression:** True physical holes and torn edges on the historical papyrus generate sharp contrast boundaries that standard segmentations frequently misinterpret as ink lines. We compute a structural buffer by applying a 7×7 pixels morphological dilation to the background gap mask, which is subsequently used to strictly zero-out any hallucinated boundary noise.

2. **Semantic-Guided Contours:** To extract sharp, crisp letter contours without capturing the underlying continuous plant fibers, a relaxed neural stencil (representing token probabilities $P > 0.55$) is fused via a logical AND operation with a localized, adaptive contrast threshold optimized at $C = 4$.
3. **Morphological Reconnection:** Damaged characters and faded text strokes often appear structurally fractured after binarization. A localized 4×4 morphological closing operation is applied to the fused binary matrix. This operation acts as an automated structural glue, successfully reconnecting disjointed vertical and horizontal strokes of individual Greek characters without bridging unrelated background noise.

Chapter Summary: The 2D masking framework contrasts a local Sauvola filtering pipeline against a frozen Vision Foundation Model (RADIO). By upscaling patches 4x, the transformer tokens map to 4x4 pixel zones. Combined with Hard Negative Mining, explicit pixel-level offline augmentations, and a custom MLP head optimized via a weighted hybrid loss, the deep learning approach replaces fragile classical heuristics with robust semantic feature extractors.

CHAPTER 3

RESULTS

To evaluate the efficacy of our 2D masking framework, we compared the advanced mathematical baseline against the fine-tuned RADIO Vision Foundation Model. The evaluation was conducted using a rigorous Leave-One-Out Cross-Validation (LOOCV) protocol. To ensure a perfectly matched, unbiased one-to-one comparison, the metrics for both the classical baseline and the deep learning architecture were computed and averaged strictly across the designated validation subset of $N = 7$ annotated fragments (Pratikakis et al. 2019).

3.1 EVALUATION METRICS

Given the severe class imbalance inherent to ancient manuscripts—where the vast majority of the image is empty papyrus substrate and only a microscopic fraction is ink—standard global accuracy is an inadequate measure. Instead, we utilized strict pixel-wise classification metrics:

- **Intersection over Union (IoU):** Serves as a strict measure of spatial alignment and contour accuracy between the predicted ink mask and the manual ground truth.
- **Dice Score (F1-Score):** Effectively balances the penalty for false positives and false negatives, making it the gold standard for sparse semantic segmentation tasks.
- **Precision and Recall:** High precision indicates the model avoids "hallucinating" ink in noisy background regions (such as dark carbonized horizontal fibers), while high recall is critical in paleography to ensure faint, degraded text strokes are not accidentally erased.

3.2 QUANTITATIVE PERFORMANCE

Prior to comparison, the mathematical baseline’s hyperparameters (such as the Sauvola threshold window and morphological kernel sizes) were empirically optimized on the validation set to provide the strongest possible classical benchmark. Despite this optimization, as presented in Table 3.1, the fine-tuned RADIO model vastly outperforms the classical pipeline across all tracked metrics.

Model Pipeline	Mean IoU $\uparrow$	Mean Dice $\uparrow$	Mean Precision $\uparrow$	Mean Recall $\uparrow$
Mathematical Baseline	0.2515 ± 0.1366	0.3814 ± 0.1893	0.6327 ± 0.2994	0.2943 ± 0.1646
RADIO VFM (LOOCV)	0.8144 ± 0.0203	0.8976 ± 0.0123	0.8301 ± 0.0172	0.9772 ± 0.0084

TABLE 3.1

Quantitative Performance on the $N = 7$ validation subset, comparing the optimized classical heuristics and the fine-tuned deep learning approach. Values represent the mean $\pm$ standard deviation across the LOOCV folds.

The mathematical baseline suffers from critically low Recall (0.2943); its reliance on sharp contrast boundaries and horizontal morphological scalpels inadvertently erases faint ink strokes and valid characters. By leveraging dense visual semantics rather than isolated pixel intensities, the deep learning architecture achieves an outstanding 0.9772 Recall while more than tripling the Mean IoU (0.8144). The significantly lower standard deviation in the VFM results (± 0.0203 for IoU versus ± 0.1366 for the baseline) also demonstrates a much higher generalization stability across vastly different fragments.

3.3 QUALITATIVE ANALYSIS

To evaluate the performance discrepancy between classical morphology and semantic deep learning, Figure 3.1 illustrates the extraction process on the Duke Papyrus Archive (*Papyrus P.Duk.inv. 765r* 2005). Each panel displays three stages: (1) the raw fragment, (2) the extracted ink mask overlaid in red, and (3) the resulting binary mask.



FIGURE 3.1

Qualitative comparison of ink extraction performance on fragment 765r. Each panel displays (1) the raw papyrus fragment, (2) the semantic ink overlay, and (3) the final binary mask.

Conversely, our proposed pipeline acts as a highly selective semantic stencil. By leveraging the frozen RADIO backbone, the model demonstrates significantly higher fidelity to the original ductus. It success-

Chapter Summary: Evaluated strictly over an identical $N = 7$ validation subset, the fine-tuned

CHAPTER 4

DISCUSSION

The experimental results validate the core hypothesis of this semester project: establishing a reliable 2D masking pipeline is a prerequisite for generating high-fidelity synthetic data, and deep learning architectures significantly outperform classical heuristics in this domain.

4.1 OVERCOMING NAIVE LIMITATIONS

The stark contrast in performance—specifically the leap in Mean Recall from 29.43% to 97.72%—highlights the limitations of naive baselines on historical substrates. The naive baseline, primarily utilizing Sauvola thresholding (Sauvola and Pietikäinen 2000) and manual morphology, operates on localized intensity and fixed geometric priors.

Beyond their susceptibility to noise, these deterministic approaches suffer from a critical lack of generalization. An optimal morphological parameter set—such as the structuring element size for fiber suppression or the specific k -factor for Sauvola thresholding—is highly sensitive to the specific illumination, ink density, and degradation pattern of a single fragment. Consequently, a heuristic that performs well on one fragment often fails catastrophically on another, necessitating tedious, manual parameter tuning for every new scan.

In contrast, the fine-tuned RADIO Vision Foundation Model (Ranzinger et al. 2024) operates on learned semantic visual descriptors. By mapping spatial tokens to localized areas, the network internalized the morphological signature of ancient Greek characters. This allows the framework to generalize across diverse papyrus collections, successfully prioritizing structural continuity over superficial contrast without requiring fragment-specific heuristic tuning.

4.2 CHALLENGES AND ROBUSTNESS

Despite the high semantic fidelity of the production pipeline, performance remains dependent on input quality. While the post-processing step effectively suppresses the majority of peripheral noise, extreme structural degradation or edge-blurring on specific optical scans can still manifest as minor artifacts in the binary masks. These edge cases highlight the necessity of maintaining robust post-processing heuristics alongside deep learning inference, rather than relying exclusively on a monolithic architecture.

4.3 LARGE-SCALE DEPLOYMENT

With the architectural robustness validated through cross-validation, the framework transitioned from evaluation to large-scale production. To maximize the semantic vocabulary of the model before deployment, a final production network was trained utilizing 100% of the available high-resolution annotated fragments ($N = 10$).

This final, highly robust 2D segmentation framework was then deployed onto optical scans sourced from collections external to the Vesuvius Challenge, notably the Duke Papyrus Archive (David M. Rubenstein Rare Book & Manuscript Library 2026) and the Oslo collection (University of Oslo Library 2026). By running inference across these extensive libraries, the model is actively generating hundreds of crisp, binary ground truth masks. This automated pipeline effectively solves the initial 2D masking bottleneck, transforming raw optical scans into a massive, standardized dataset of isolated ancient text.

4.4 PATHWAY TO 3D VOLUMETRIC SYNTHESIS AND BROADER IMPACT

The successful generation of this 2D dataset directly serves the original overarching goal of this project: addressing the critical 3D data scarcity within the Vesuvius Challenge.

As demonstrated by our preliminary experiments with 3D generation architectures such as Masked Autoencoders (MAE), early attempts at volumetric generation were fundamentally bottlenecked by noisy, heuristic-based 2D annotations. The masks generated by our VFM pipeline now provide a flawless foundational layer. Building upon prior concepts for synthetic generation (Parker 2023), future work will focus on integrating this new dataset into volumetric algorithms. By projecting these highly accurate 2D ink layouts into synthetic 3D space, researchers will be able to artificially simulate the morphological signature of carbonized scrolls, providing the Vesuvius Challenge community with the massive volumetric training data required to decipher the hidden texts of Herculaneum.

Furthermore, beyond the immediate scope of 3D synthesis, the open-source publication of this large-scale 2D dataset holds significant standalone value. It provides the broader computational paleography and digital humanities communities with an unprecedented resource of pixel-perfect ground truth. This empowers other researchers and institutions to benchmark novel segmentation architectures, study ancient Greek ductus, and tackle diverse epigraphic challenges across other historical collections.

Chapter Summary: The VFM’s semantic understanding inherently bypasses the fragility of classical pixel-level filters. Validated by exceptional Recall scores, the model has been deployed on external archives to mass-produce 2D masks. This newly generated dataset completely resolves the 2D bottleneck, laying the essential groundwork for robust 3D synthetic data generation in the Vesuvius Challenge while offering a highly valuable standalone resource for the broader scientific community.

CHAPTER 5

CONCLUSION

The Vesuvius Challenge represents one of the most formidable computational archaeology initiatives of our time, yet it remains fundamentally constrained by a lack of high-fidelity volumetric training data (Vesuvius Challenge 2023). Through this project, we explored a specific methodology to address this deficit via synthetic data generation. We demonstrated that this approach first requires solving a foundational 2D masking bottleneck, as our preliminary experiments showed that simplistic 2D-to-3D padding techniques are fundamentally insufficient to capture the deep, complex physical penetration of carbon ink into papyrus fibers.

By systematically comparing an advanced, handcrafted mathematical filtering pipeline (Sauvola and Pietikäinen 2000) against a modern deep learning architecture, we highlighted the profound limitations of training-free, heuristic-based methods in extreme low-signal historical domains. Deterministic algorithms inevitably fail to capture the complex, faded morphology of ancient ink, resulting in unacceptable levels of data loss.

In contrast, our custom pipeline—leveraging the dense semantic descriptors of the frozen RADIO Vision Foundation Model (Ranzinger et al. 2024) paired with a localized MLP head—achieved a paradigm shift in performance. Validated on a strict LOOCV subset, the deep learning model inherently understood the geometric structure of Greek characters, achieving an outstanding Mean Recall of 97.72% and a Mean IoU of 81.44%, proving entirely robust to severe substrate degradation.

Ultimately, this project successfully transitioned from architectural validation to large-scale production. By deploying the fine-tuned framework onto unseen external datasets, specifically the Duke Papyrus Archive (David M. Rubenstein Rare Book & Manuscript Library 2026) and the Oslo collection (University of Oslo Library 2026), we established a fully automated pipeline capable of generating hundreds of binary ground truth masks. This massive 2D dataset effectively resolves the masking bottleneck for our generation pipeline, laying the flawless foundational groundwork required to power future 3D volumetric synthesis algorithms and accelerate the global effort to decipher the scrolls of Herculaneum.

AI USAGE APPENDIX

During the development of this project, a Large Language Model (Gemini) was utilized as a technical assistant. It was primarily employed for drafting PyTorch training boilerplates, debugging CPU-to-GPU data streaming bottlenecks, and streamlining the academic formatting of this report. All methodological decisions, architectural choices, hyperparameter tunings, and final data analyses were strictly conducted, verified, and validated by the author.

Chapter Summary: The deep learning framework definitively solved the 2D masking bottleneck inherent to our synthetic generation approach, outperforming training-free mathematical baselines by a massive margin. Deployed on the Duke and Oslo archives, the resulting large-scale 2D dataset provides the ground truth necessary to finally unlock 3D synthetic data generation for the Vesuvius Challenge and benefit the broader paleography community.

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